

# Research Statement

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My research focuses on knowledge representation and reasoning, in particular, on reasoning about probabilistic and higher-order beliefs. I aim to develop formal and neuro-symbolic methods for epistemic reasoning in multi-agent systems.

During my Ph.D. study, my research can be categorized into four lines of work:

- **Probabilistic reasoning:** Expressive logical frameworks that integrate probability theory with possible-world semantics, enabling reasoning about probability distributions in both static and dynamic settings [2, 5, 1].
- **Modal theorem proving:** A translation-based theorem prover for multi-agent modal logics with introspection, which preserves the decidability and supports a first-order extension [3].
- **Planning:** A syntactic approach for multi-agent epistemic planning. By bounding the Theory of Mind (ToM) order, it preserves decidability while maintaining most of the expressiveness of existing (undecidable) approaches. (*recently accepted to AAAI'26 for an oral presentation*).
- **Neurosymbolic methods** — Integrating formal reasoning with vision-language models (VLMs) and large language model (LLMs) for explainable and trustworthy image anomaly detection [4].

In the next stage of my research, I plan to work along two closely related directions:

- I will extend my recent work on multi-agent epistemic planning to broader reasoning tasks, including logical programming for higher-order beliefs, strategy synthesis, and epistemic protocol synthesis.
- I plan to apply the above theoretical frameworks to practical, real-world tasks, such as social deduction games and human-agent interaction. To achieve this, integrating subsymbolic methods will be important. Some possible directions are
  1. reasoning with probabilistic defaults provided by pre-trained neural networks,
  2. RL-based heuristic search for epistemic planning, and
  3. bridging formal reasoning with natural language via LLM-based semantic parsers.
- In response to integrating agents into society, I plan to explore the formalization of concepts in social science, such as Jürgen Habermas' Universalization Principle of moral norm, or the theory of communicative actions. This could be achieved by introducing additional modalities and epistemic variants of the Situation Calculus.

So far, I have collaborated with colleagues from Germany (Aachen), the UK (Edinburgh), China (Nanjing), and Canada (Regina), and I look forward to further expanding these international and interdisciplinary collaborations.

## References

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